Overview of
Prediction Markets

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CBER CtCe, Columbia University, October 29, 2026

Outline

PART I

Conceptual Framework

What is a prediction market?

PART II

Research Overview

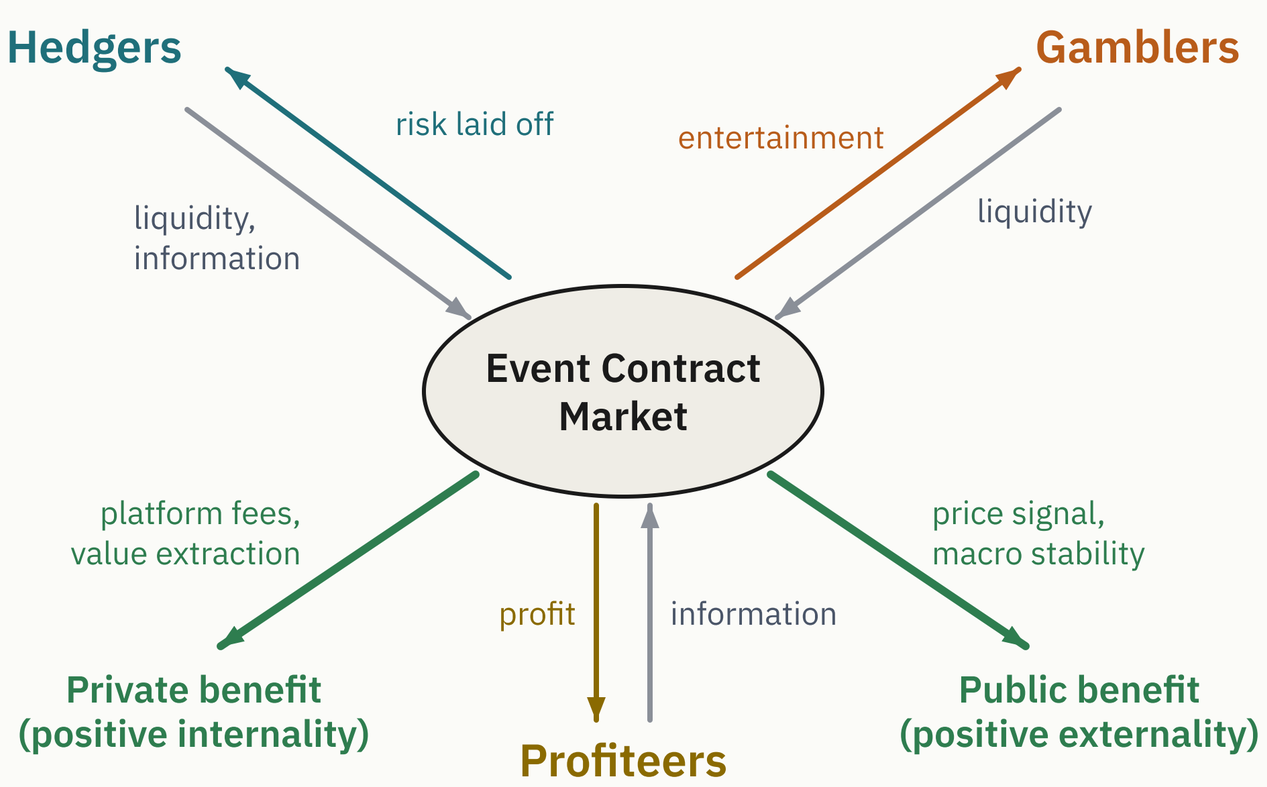
Accuracy, bias, and analysis of offshore prediction markets.

PART III

Practical Matters

What Parts I and II say about regulatory and legal questions.

Prediction markets as a complex system



Interesting markets attract Gamblers

who supply liquidity, price distortions

Risky events attract Hedgers

who supply liquidity and pay a premium to lay off risk

Profit opportunities attract Profiteers

who profit from mispricing, correct mispricings

Which events attract whom?

Event	Interesting	Risky	Profitable
Presidential election	●●●	●●●	●●●
Papal election	●●○	○○○	●○○
Super Bowl	●●●	●○○	●●●
Pre-season game	●○○	○○○	●○○
Box office	●●○	●●○	●●○
Rotten Tomatoes score	●○○	○○○	●○○

Perspective: Public and private benefits/harms are emergent properties of a complex system.

PART I · FUNDAMENTAL CONCEPTS

Price constrains probability.

Crane (2024). *The Margin of Inefficiency in Prediction Markets.*
Crane. *Prediction Markets* (book, forthcoming), Chapters 3 and 6.

1. The Core Principle and the Fundamental Theorem

CORE PRINCIPLE OF TRADING

For an event contract C,

buy C if and only if Price(C) < Value(C),

sell C if and only if Price(C) > Value(C).

Price is what something costs.
Value is what something is worth.
Price ≠ Value.

Value(C) = **intrinsic** value + **extrinsic** value.

Profiteers trade on intrinsic value.

Hedgers trade on extrinsic value.

Gamblers trade on extrinsic value.

A trade is an agreement to disagree on value.

FUNDAMENTAL THEOREM OF PREDICTION MARKETS

A market is **efficient** if and only if every quoted contract C satisfies

$$\text{Price}(C) \geq \text{IntrinsicValue}(C).$$

Implies: Price(Yes) ≥ Prob(Yes), Price(No) ≥ Prob(No).

Price constrains probability

Per-contract fee f: Price(Yes) = Ask(Yes) + f and Price(No) = 1 − Bid(Yes) + f, so

$$\text{Bid(Yes)} - f \leq \text{Prob(Yes)} \leq \text{Ask(Yes)} + f$$

Order book	Yes	No
Best ask (bid on Yes is $0.59)	$0.61	$0.41
Fee per contract	$0.02	$0.02
Effective price	$0.63	$0.43
Efficient if	**57% ≤ Prob(Yes) ≤ 63%**	

Crane, *Prediction Markets* (book, forthcoming), Chapters 1 and 3.

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2. Interval of inefficiency

INTERVAL OF INEFFICIENCY (II)

The probabilities for which **neither** Yes nor No is profitable after friction.

Consensus price = midpoint

Margin of inefficiency (MOI) = half the width

$$II = Consensus \pm MOI$$

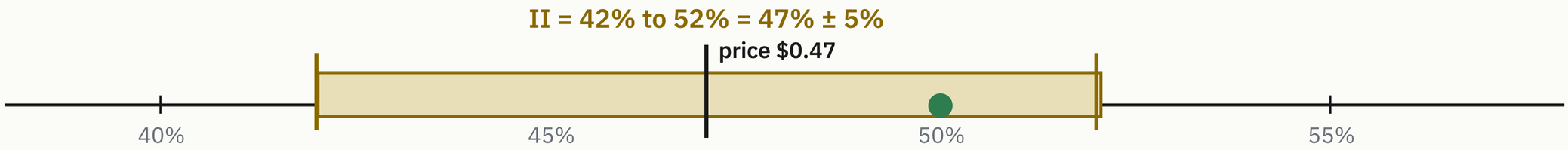
Relation to margin of error

Poll	51% ± 3%	sampling error, refers to vote share
Market	47% ± 5%	friction, refers to probability

Endogenous friction: fees, spread, tick, settlement date.

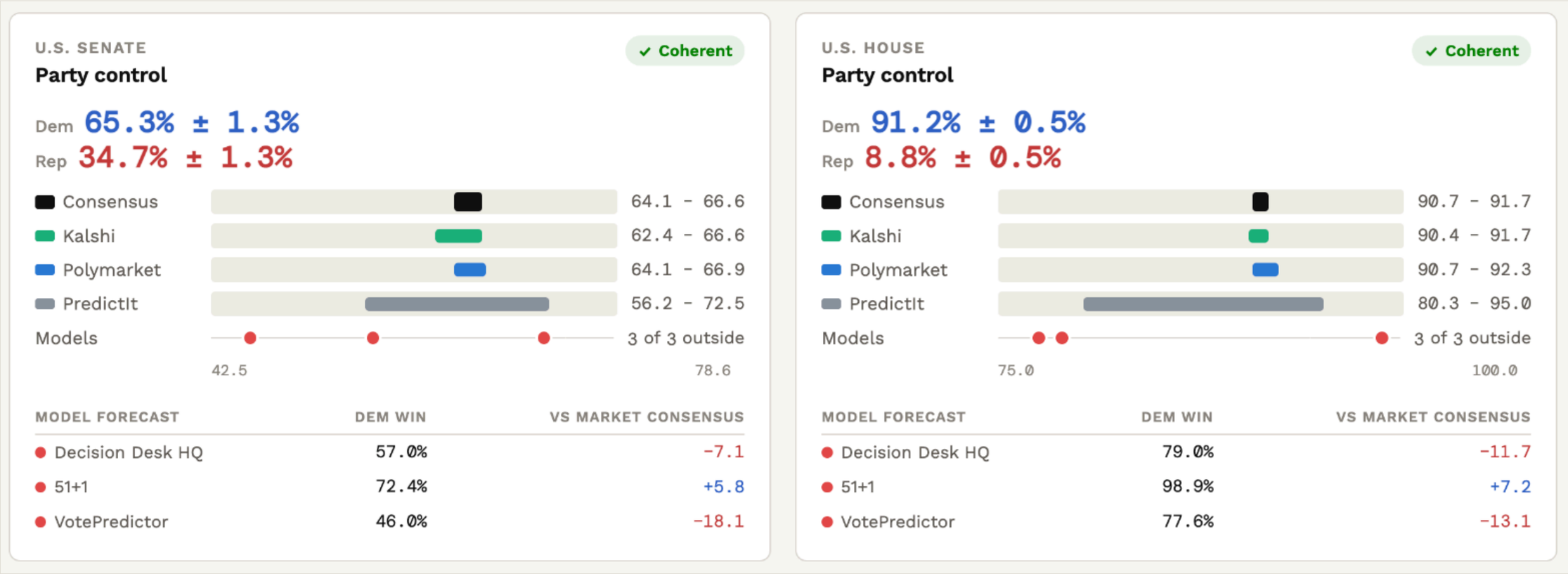
Exogenous friction: cost of capital, liquidity risk, access limits.

Worked example	Yes trades at \$0.47, No at \$0.53. Flat fee \$0.05 per share. Assume true probability is 50%.	
Buy Yes at \$0.47	0.47 + 0.05 = 0.52 → Prob(Yes) ≤ 52%	EV = 0.50 × \$1 − \$0.52 = **−\$0.02** per share
Buy No at \$0.53	0.53 + 0.05 = 0.58 → Prob(Yes) ≥ 42%	EV = 0.50 × \$1 − \$0.58 = **−\$0.08** per share
Interval of inefficiency	**42% to 52% = 47% ± 5%**	**Price ≠ probability, yet neither side is worth buying.**



Crane (2024), *The Margin of Inefficiency in Prediction Markets*; Crane, *Prediction Markets* (book, forthcoming), Chapter 6.

3. The interval in practice: harrycrane.com/elections



U.S. Senate and House control, 2026 midterms. Screenshot October 26, 2026, 5:03 PM ET.

Methodology: Crane (2026), *The Margin of Inefficiency*.

PART II · ACCURACY

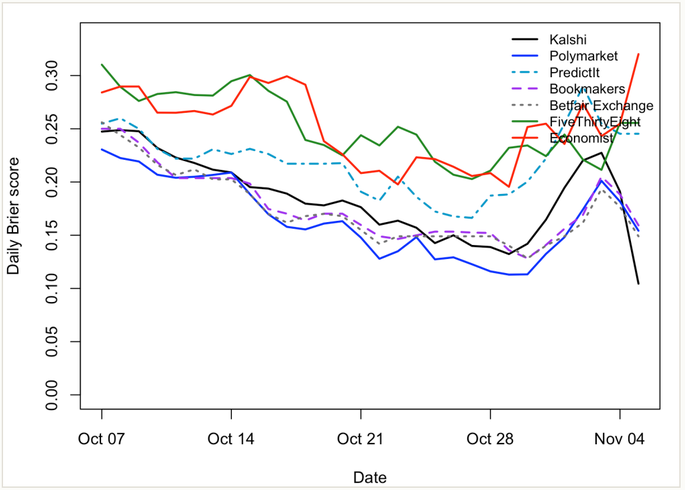
Markets > Models

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Crane and Strumpf (2026). *Diversity, Efficiency, and Incentives in Political Prediction Markets.* Working paper.

Accuracy: market-based scoring

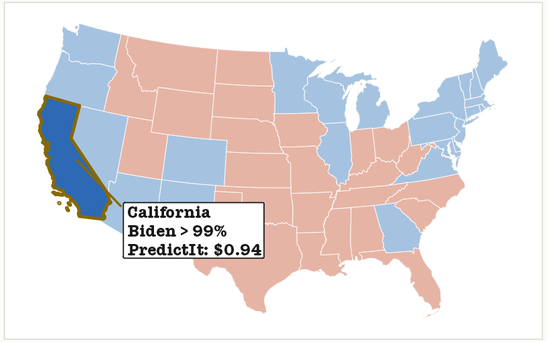
2024 Election Accuracy

- **Markets vs. models.** PredictIt, Kalshi, Polymarket, Betfair, bookmakers vs. 538, The Economist.
- **Market-based scoring.** If a model beats the market, trading on it should be profitable.



2024 daily Brier score, national race: every market beat every model.

A known distortion



2020 results by state. California highlighted.

$0.94

Election day 2020: Biden to win California on PredictIt.

Why?

- Deposit $850 (the cap) to buy 904 shares. Win $904; after 10% commission on profit ($5.40) and 5% withdrawal fee, $854: a $4 profit.
- The price reflects rational understanding of **friction**, not market opinion about California.

Markets tend to be accurate, but market design and structure can lead to bias or distortions.

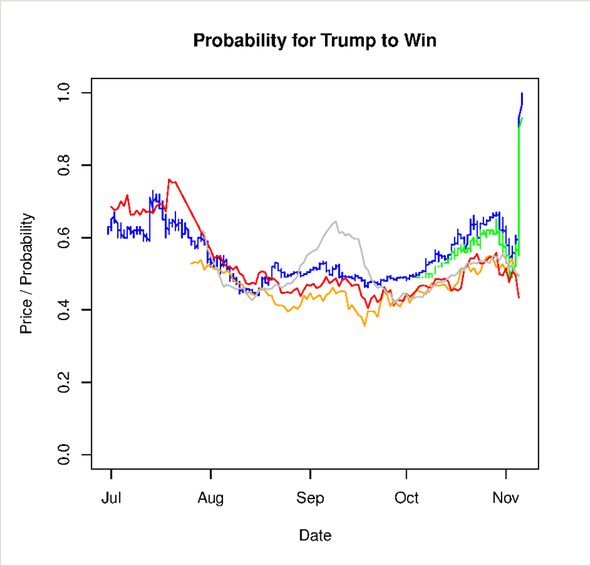
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PART II · MODELS

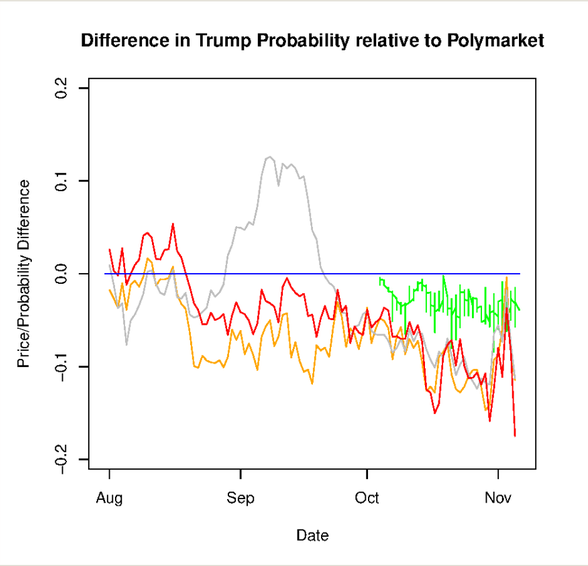
Case study: models behaving badly.

Crane and Strumpf (2026). *Diversity, Efficiency, and Incentives in Political Prediction Markets.* Working paper.

Model hallucination



Implied Trump probability, July to November 2024.



Same series, relative to Polymarket.

Post-convention bounce

- Model forecast (gray) assumes a "post-convention bounce" adjustment.
- Polls did not move toward Harris -> flat polling read as a pro-Trump signal.

Without alternative/independent forecasts, model assumptions are hidden and error goes undetected.

PART II · WHO TRADES

Bias in Political Prediction Markets.

Crane and Strumpf (2026). *Diversity, Efficiency, and Incentives in Political Prediction Markets.* Working paper.
Rhode and Strumpf (2004). *Historical Presidential Betting Markets.* Journal of Economic Perspectives 18(2).

Bias in Political Prediction Markets

Traders are not representative of voters.

Group	Traders	Volume	Electorate
Male	85.8%	96.9%	47%
Under 30	51.5%	42.1%	12.9%
White*	89.6%	97.1%	64.3%

* Among the 45% of users with a confident BISG classification.

- The modal trader: a white male under 35.
- Dollars came from zip codes that voted **28% Republican**. An expressive market would have leaned *Harris*.

Incentives, not demographics

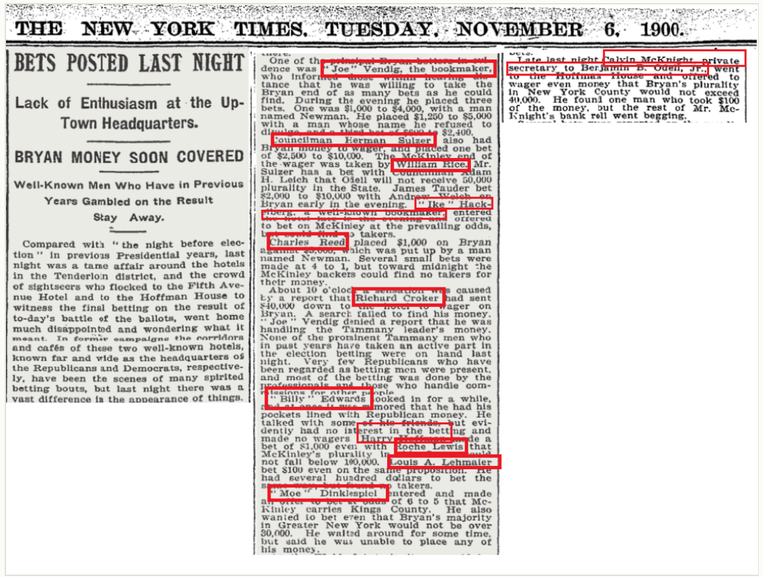
Trades	Users	% long Trump	Partisan gap (pts)	ROI
1	25,785	27	6.8	−56%
2–10	113,830	56	6.9	−10%
11–100	48,258	59	4.5	−10%
101–5,000	10,150	68	2.9	−2%
>5,000	**420**	64	**−0.9**	+1.5%

Kalshi traders by trade count, main presidential market, Oct. 4 to Nov. 5, 2024. Partisan gap = % of dollars long Republican from Republican-attributed traders minus Democrat-attributed traders, attributed by the 2024 vote of the trader's home zip code; dollar-weighted, all party-classifiable election markets.

> *"I never realised that to be a jockey you had to be a horse first."*
>
> Arrigo Sacchi, AC Milan manager, to critics who noted he had never played professionally.

Individual participants are biased, but the market removes the bias.

The same pattern in 1900 and 2016



THE NEW YORK TIMES, TUESDAY, NOVEMBER 6, 1900.

BETS POSTED LAST NIGHT

Lack of Enthusiasm at the Up-Town Headquarters.

BRYAN MONEY SOON COVERED

Well-Known Men Who Have in Previous Years Gambled on the Result Stay Away.

Compared with "the night before election" in previous Presidential years, last night was a tame affair around the hotels in the Tenderloin district, and the crowd of sightseers who flocked to the Fifth Avenue Hotel and to the Hoffman House to witness the final betting on the result of to-day's battle of the ballots, went home much disappointed and wondering what it meant. In former campaigns the corridors and cafés of these two well-known hotels, known far and wide as the headquarters of the Republicans and Democrats, respectively, have been the scenes of many spirited betting bouts, but last night there was a vast difference in the appearance of things.

One of the [illegible] evidence was "Joe" Vendig, the bookmaker, who informed those within hearing distance that he was willing to take the Bryan end of as many bets as he could find. During the evening he placed three bets. One was $1,000 to $4,000, with a man named Newman. He placed $1,250 to $5,000 with a man whose name he refused to disclose, making $2,400.

Councilman Herman Sulzer also had Bryan money to wager, and placed one bet of $2,000 to $10,000. The McKinley end of the wager was taken by William Rice. Mr. Sulzer has a bet with Councilman Adam H. Letch that Odell will not receive 50,000 plurality in the State. James Tauder bet $2,000 to $10,000 with Andrew Welch on Bryan early in the evening. "Ike" Hackett, a well-known bookmaker, entered [illegible] evening and offered to bet on McKinley at the prevailing odds, but found no takers.

Charles Reed placed $1,000 on Bryan at the Hoffman House, which was put up by a man named Newman. Several small bets were made at 4 to 1, but toward midnight the McKinley backers could find no takers for their money.

About 10 o'clock a sensation was caused by a report that Richard Croker had sent $40,000 down to the front to wager on Bryan. A search failed to find his money.

"Joe" Vendig denied a report that he was handling the Tammany leader's money. None of the prominent Tammany men who in past years have taken an active part in the election betting were on hand last night. Very few Republicans who have been regarded as betting men were present, and most of the betting was done by the professionals and those who handle commissions for other people.

"Billy" Edwards looked in for a while, but apparently was more interested that he had his pockets lined with Republican money. He talked with some of his friends, but evidently had no interest in the betting and made no wagers. Harry Bodine made a bet of $1,000 even with Roche Lewis that McKinley's plurality in [illegible] would not fall below 100,000. Louis A. Lehmaier bet $100 even on the same proposition. He had several hundred dollars to bet the same way, but found no takers.

"Moe" Dinkiespiel entered and made a wager of 6 to 5 that McKinley carries Kings County. He also wanted to bet even that Bryan's majority in Greater New York would not be over 30,000. He waited around for some time, but said he was unable to place any of his money.

Late last night Calvin McKnight, private secretary to Benjamin B. Odell, Jr., went to the Hoffman House and offered to wager even money that Bryan's plurality in New York County would not exceed 40,000. He found one man who took $100 of the money, but the rest of Mr. McKnight's bank roll went begging.

New York Times, Nov. 6, 1900. Boxed passages report the odds and name bettors with their stakes.

New York curb market, 1900

- 80 named bettors matched to the 1900 Census: 44% bankers and brokers (1% of the city), older, married, native-born.
- Unrepresentative in every dimension, and accurate: the Wall Street odds called presidential races for decades (Rhode and Strumpf 2004).

PredictIt, 2016

- Registered party known. Positions followed party more closely than on Kalshi in 2024.
- Possible explanation: the $850 cap stops profit-seekers from correcting prices, so expressive trading leaks into the price.

Crane and Strumpf (2026). *Diversity, Efficiency, and Incentives in Political Prediction Markets.* Working paper. Rhode and Strumpf (2004), *Journal of Economic Perspectives*.

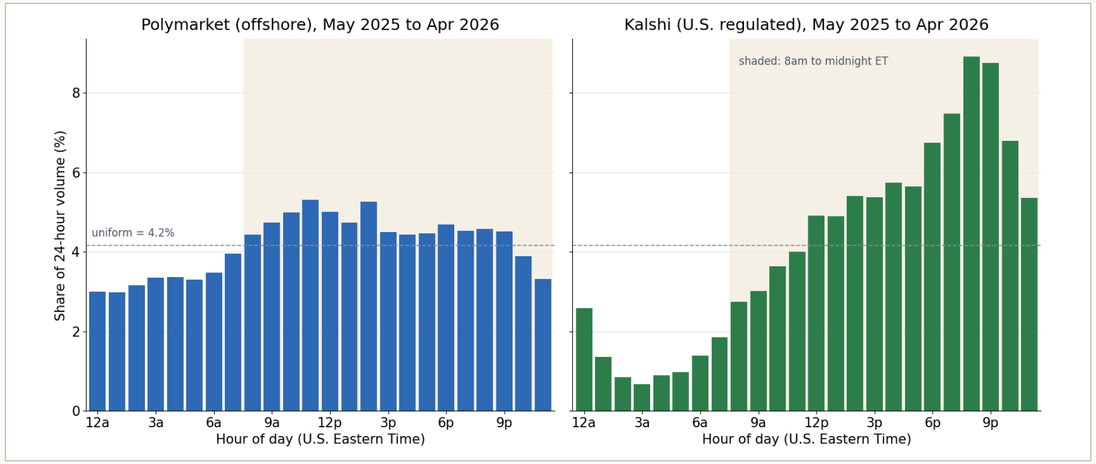
PART II · OFFSHORE

Prohibition is not prevention.

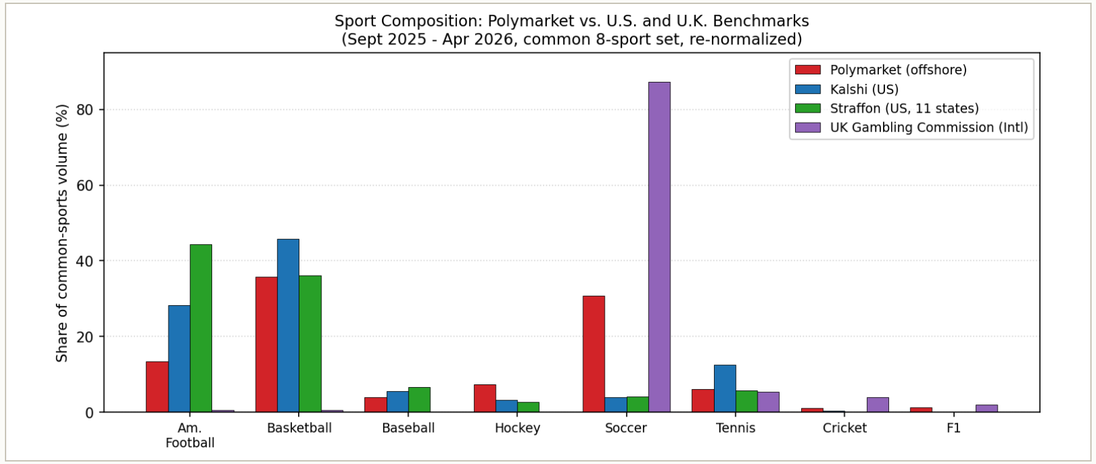
Crane (2026). *Estimating U.S. User Activity in Offshore Prediction Markets.* Research Brief and Technical Report, June 2026.

Offshore activity from U.S. users

Regulated exchanges traded about $74B and offshore platforms $85B in the twelve months to April 2026.



Distribution of trading time. Kalshi volume follows the U.S. day and evening; Polymarket is flatter, a mix of U.S. and overseas hours.



Distribution of sports volume. U.S. money goes to basketball and football, international money to soccer; Polymarket's mix is between the two.

~30%

of Polymarket (offshore) user activity estimated to come from U.S.-based users

$11–34B

of offshore volume, May 2025 to April 2026, originating in the United States

Regulatory lesson: prohibition is not prevention. Banning a contract from regulated exchanges moves the activity offshore.

PART III

Practical Matters

Implications for market design and prediction market regulation.

Crane (2024, 2025, 2026). *Futures and Derivatives Law Report* 44(8), 45(8), 46(9).

Controversial markets and market integrity

1. Gaming

- 2024 Commission: politics and sports are gaming.
- 2026 proposal: gaming = involving a game.
- Proposal: gaming = **frivolous risk-taking** (an event that exists only to be bet on).

2. War and terrorism

- **Hedging** for airlines, shippers, insurers, families in conflict zones.
- **Information** when official sources unreliable.
- Strait of Hormuz: Kalshi settles on a 90% drop in shipping traffic, Polymarket on 80%.
- Oil futures and traditional markets move on the same conflict.

3. Sports: hedging use case

- Gallery Furniture, $50M hedge on the 2022 World Series.
- LSU coaching incentives, ticket revenue, broadcast rights.

Market integrity: common misconceptions

French Whale, 2024. Large Trump position built on private polling. Suspected as manipulation or insider trading. Later reporting suggested that Freddi conducted his own polls and believed Trump's chances higher than public polling.

Information advantage is not manipulation or insider trading.

Nobel Peace Prize, 2025. Bet placed hours before the announcement. Suspected as insider trading or "espionage". Later revealed as an accidental information leak by Nobel Committee.

Markets incentivize information discovery.

Maduro raid, 2026. A U.S. soldier traded on classified knowledge of a military operation on Polymarket (offshore).

Markets expose bad behavior easily.

Khamenei out of office, 2026. Paid $1.00 on Polymarket (offshore) and $0.09 on Kalshi (U.S.) because of a death carve-out in the contract.

Hedging and information value of market diminished by carve outs and edge cases.

Questions	1. How to quantify the public benefit and potential harm of any given market?	2. Where is the line between information advantage and manipulation or insider trading?	3. How should contracts be designed so that resolution stays faithful to the underlying event?

Crane (2024, 2025, 2026). *FDLR* 44(8), 45(8), 46(9). Comment letters on RIN 3038-AF14 and RIN 3038-AF65.

Sources

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Public comment and advisory work

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Crane, H. (2026). Comment on Advance Notice of Proposed Rulemaking: Prediction Markets (RIN 3038-AF65). Apr. 30, 2026.

CFTC Innovation Advisory Committee. Member; remarks at the inaugural meeting, Aug. 20, 2026.

Crane, H. (2026). Comment on the Inaugural Meeting of the CFTC Innovation Advisory Committee. Submitted Aug. 26, 2026.

Public and private benefits and harms are emergent properties of a complex system.